

# Chinook Music Store

## SQL Business Questions Project

This project frames SQL practice as a set of real business questions asked by different stakeholders — Sales, Finance, Marketing, and Management — using the Chinook sample database (a digital music store with customers, invoices, tracks, artists, and employees). For each question, the SQL query used is recorded along with the answer or insight it produced.

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
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| <b>Dataset</b>     | Chinook sample database (chinook-database, GitHub)                    |
| <b>Tables used</b> | Customer, Invoice, InvoiceLine, Track, Album, Artist, Genre, Employee |

### Tier 1 — Single Table (Sales team asks)

#### Q1. How many total customers do we have, broken down by country?

*Hint: GROUP BY country, COUNT(\*)*

##### SQL Query

```
SELECT Country, COUNT(*) AS Total_Customers
FROM Customer
GROUP BY Country;
```

##### Answer / Insight

| Country        | Customers |
|----------------|-----------|
| USA            | 13        |
| Canada         | 8         |
| Brazil         | 5         |
| France         | 5         |
| Germany        | 4         |
| United Kingdom | 3         |
| Czech Republic | 2         |
| Portugal       | 2         |
| India          | 2         |
| Belgium        | 1         |
| Hungary        | 1         |
| Spain          | 1         |
| Denmark        | 1         |
| Ireland        | 1         |
| Sweden         | 1         |

| Country     | Customers |
|-------------|-----------|
| Italy       | 1         |
| Norway      | 1         |
| Netherlands | 1         |
| Australia   | 1         |
| Poland      | 1         |
| Argentina   | 1         |
| Austria     | 1         |
| Finland     | 1         |
| Chile       | 1         |

The store's customer base is concentrated in North America and Western Europe. The USA (13), Canada (8), Brazil (5), and France (5) together account for roughly half of all 59 customers, while most other countries contribute just one or two.

## Q2. What are the 10 most expensive tracks we sell?

Hint: ORDER BY UnitPrice DESC, LIMIT 10

### SQL Query

```
SELECT TrackId, Name, UnitPrice
FROM Track
ORDER BY UnitPrice DESC
LIMIT 10;
```

### Answer / Insight

| Track ID | Name                                   | Unit Price |
|----------|----------------------------------------|------------|
| 2822     | Exodus, Pt. 2                          | 1.99       |
| 2828     | The Passage                            | 1.99       |
| 2825     | A Measure of Salvation                 | 1.99       |
| 2823     | Collaborators                          | 1.99       |
| 2827     | Unfinished Business                    | 1.99       |
| 2821     | Exodus, Pt. 1                          | 1.99       |
| 2826     | Hero                                   | 1.99       |
| 2824     | Torn                                   | 1.99       |
| 2820     | Occupation / Precipice                 | 1.99       |
| 2819     | Battlestar Galactica: The Story So Far | 1.99       |

All ten most expensive tracks are priced identically at \$1.99, and every one of them belongs to the 'Battlestar Galactica' catalog — the store's premium pricing tier appears to be tied to this one title rather than spread across artists.

## Q3. Which customers are missing a Fax number or a State value?

Hint: WHERE Fax IS NULL OR State IS NULL

### SQL Query

```
SELECT CustomerId, FirstName, LastName, Fax, State
FROM Customer
WHERE Fax IS NULL
```

OR State IS NULL;

### Answer / Insight

| ID | First Name | Last Name   | Fax              | State  |
|----|------------|-------------|------------------|--------|
| 2  | Leonie     | Köhler      | NULL             | NULL   |
| 3  | François   | Tremblay    | NULL             | QC     |
| 4  | Björn      | Hansen      | NULL             | NULL   |
| 5  | František  | Wichterlová | +420 2 4172 5555 | NULL   |
| 6  | Helena     | Holý        | NULL             | NULL   |
| 7  | Astrid     | Gruber      | NULL             | NULL   |
| 8  | Daan       | Peeters     | NULL             | NULL   |
| 9  | Kara       | Nielsen     | NULL             | NULL   |
| 20 | Dan        | Miller      | NULL             | CA     |
| 21 | Kathy      | Chase       | NULL             | NV     |
| 22 | Heather    | Leacock     | NULL             | FL     |
| 23 | John       | Gordon      | NULL             | MA     |
| 24 | Frank      | Ralston     | NULL             | IL     |
| 25 | Victor     | Stevens     | NULL             | WI     |
| 26 | Richard    | Cunningham  | NULL             | TX     |
| 27 | Patrick    | Gray        | NULL             | AZ     |
| 28 | Julia      | Barnett     | NULL             | UT     |
| 29 | Robert     | Brown       | NULL             | ON     |
| 30 | Edward     | Francis     | NULL             | ON     |
| 31 | Martha     | Silk        | NULL             | NS     |
| 32 | Aaron      | Mitchell    | NULL             | MB     |
| 33 | Ellie      | Sullivan    | NULL             | NT     |
| 34 | João       | Fernandes   | NULL             | NULL   |
| 35 | Madalena   | Sampaio     | NULL             | NULL   |
| 36 | Hannah     | Schneider   | NULL             | NULL   |
| 37 | Fynn       | Zimmermann  | NULL             | NULL   |
| 38 | Niklas     | Schröder    | NULL             | NULL   |
| 39 | Camille    | Bernard     | NULL             | NULL   |
| 40 | Dominique  | Lefebvre    | NULL             | NULL   |
| 41 | Marc       | Dubois      | NULL             | NULL   |
| 42 | Wyatt      | Girard      | NULL             | NULL   |
| 43 | Isabelle   | Mercier     | NULL             | NULL   |
| 44 | Terhi      | Hämäläinen  | NULL             | NULL   |
| 45 | Ladislav   | Kovács      | NULL             | NULL   |
| 46 | Hugh       | O'Reilly    | NULL             | Dublin |
| 47 | Lucas      | Mancini     | NULL             | RM     |

| ID | First Name | Last Name    | Fax  | State |
|----|------------|--------------|------|-------|
| 48 | Johannes   | Van der Berg | NULL | VV    |
| 49 | Stanisław  | Wójcik       | NULL | NULL  |
| 50 | Enrique    | Muñoz        | NULL | NULL  |
| 51 | Joakim     | Johansson    | NULL | NULL  |
| 52 | Emma       | Jones        | NULL | NULL  |
| 53 | Phil       | Hughes       | NULL | NULL  |
| 54 | Steve      | Murray       | NULL | NULL  |
| 55 | Mark       | Taylor       | NULL | NSW   |
| 56 | Diego      | Gutiérrez    | NULL | NULL  |
| 57 | Luis       | Rojas        | NULL | NULL  |
| 58 | Manoj      | Pareek       | NULL | NULL  |
| 59 | Puja       | Srivastava   | NULL | NULL  |

48 of the 59 customers (over 80%) are missing a Fax number, a State value, or both — this field is effectively unused for most records and shouldn't be relied on for reporting or contact purposes.

## Tier 2 — Aggregation (Finance asks)

### Q4. What is our total revenue, and what is the average invoice value?

Hint: `SUM(Total)`, `AVG(Total)` from Invoice

#### SQL Query

```
SELECT
  SUM(Total) AS total_revenue,
  AVG(Total) AS average_invoice_value
FROM Invoice;
```

#### Answer / Insight

| Total Revenue | Average Invoice Value |
|---------------|-----------------------|
| \$2328.60     | \$5.651942            |

Total lifetime revenue is \$2,328.60 across all invoices, with a modest average order value of about \$5.65 — consistent with a store that sells individual tracks rather than bundled albums.

### Q5. Which country generates the most revenue? Rank all countries.

Hint: `GROUP BY BillingCountry`, `ORDER BY SUM(Total) DESC`

#### SQL Query

```
SELECT
  BillingCountry,
  SUM(Total) AS total_revenue
FROM Invoice
GROUP BY BillingCountry
ORDER BY SUM(Total) DESC;
```

#### Answer / Insight

| Country        | Total Revenue |
|----------------|---------------|
| USA            | \$523.06      |
| Canada         | \$303.96      |
| France         | \$195.10      |
| Brazil         | \$190.10      |
| Germany        | \$156.48      |
| United Kingdom | \$112.86      |
| Czech Republic | \$90.24       |
| Portugal       | \$77.24       |
| India          | \$75.26       |
| Chile          | \$46.62       |
| Ireland        | \$45.62       |
| Hungary        | \$45.62       |
| Austria        | \$42.62       |
| Finland        | \$41.62       |
| Netherlands    | \$40.62       |
| Norway         | \$39.62       |
| Sweden         | \$38.62       |
| Belgium        | \$37.62       |
| Australia      | \$37.62       |
| Spain          | \$37.62       |
| Denmark        | \$37.62       |
| Italy          | \$37.62       |
| Poland         | \$37.62       |
| Argentina      | \$37.62       |

The USA leads by a wide margin at \$523.06, more than 70% ahead of second-place Canada (\$303.96). Revenue roughly tracks customer count from Q1, though a few smaller markets (e.g. India, Czech Republic) punch above their customer share.

## Q6. What is our revenue per year? Is it trending up or down?

Hint: GROUP BY EXTRACT(YEAR FROM InvoiceDate)

### SQL Query

```
SELECT
  EXTRACT(YEAR FROM InvoiceDate) AS invoice_year,
  SUM(Total) AS total_revenue
FROM Invoice
GROUP BY invoice_year
ORDER BY invoice_year ASC;
```

### Answer / Insight

| Year | Total Revenue |
|------|---------------|
| 2021 | \$449.46      |
| 2022 | \$481.45      |

| Year | Total Revenue |
|------|---------------|
| 2023 | \$469.58      |
| 2024 | \$477.53      |
| 2025 | \$450.58      |

Annual revenue has been remarkably flat, holding in a narrow \$449–\$481 band from 2021 through 2025 with no clear upward or downward trend — the business is stable but not growing.

## Tier 3 — Joins (Marketing asks)

### Q7. Which genre has generated the most revenue?

Hint: JOIN InvoiceLine -> Track -> Genre, GROUP BY genre name

#### SQL Query

```
SELECT
    Genre.Name,
    SUM(InvoiceLine.UnitPrice * InvoiceLine.Quantity) AS total_revenue
FROM InvoiceLine
JOIN Track ON InvoiceLine.TrackId = Track.TrackId
JOIN Genre ON Track.GenreId = Genre.GenreId
GROUP BY Genre.Name
ORDER BY total_revenue DESC;
```

#### Answer / Insight

| Genre              | Total Revenue |
|--------------------|---------------|
| Rock               | \$826.65      |
| Latin              | \$382.14      |
| Metal              | \$261.36      |
| Alternative & Punk | \$241.56      |
| TV Shows           | \$93.53       |
| Jazz               | \$79.20       |
| Blues              | \$60.39       |
| Drama              | \$57.71       |
| R&B/Soul           | \$40.59       |
| Classical          | \$40.59       |
| Sci Fi & Fantasy   | \$39.80       |
| Reggae             | \$29.70       |
| Pop                | \$27.72       |
| Soundtrack         | \$19.80       |
| Comedy             | \$17.91       |
| Hip Hop/Rap        | \$16.83       |
| Bossa Nova         | \$14.85       |
| Alternative        | \$13.86       |
| World              | \$12.87       |

| Genre             | Total Revenue |
|-------------------|---------------|
| Science Fiction   | \$11.94       |
| Heavy Metal       | \$11.88       |
| Electronica/Dance | \$11.88       |
| Easy Listening    | \$9.90        |
| Rock And Roll     | \$5.94        |

Rock dominates with \$826.65 in revenue — more than double the next genre (Latin, \$382.14) — suggesting catalog and marketing investment should continue to favor Rock while niche genres like Rock And Roll (\$5.94) contribute little.

## Q8. Who are our top 5 customers by total spend, and which sales rep manages them?

Hint: JOIN Customer -> Employee via SupportRepId

### SQL Query

```
SELECT
    Customer.FirstName,
    Customer.LastName,
    SUM(Invoice.Total) AS total_spend,
    Employee.FirstName AS rep_first_name,
    Employee.LastName AS rep_last_name
FROM Invoice
JOIN Customer ON Invoice.CustomerId = Customer.CustomerId
JOIN Employee ON Customer.SupportRepId = Employee.EmployeeId
GROUP BY
    Customer.CustomerId, Customer.FirstName, Customer.LastName,
    Employee.EmployeeId, Employee.FirstName, Employee.LastName
ORDER BY total_spend DESC
LIMIT 5;
```

### Answer / Insight

| First Name | Last Name  | Total Spend | Rep First Name | Rep Last Name |
|------------|------------|-------------|----------------|---------------|
| Helena     | Holý       | 49.62       | Steve          | Johnson       |
| Richard    | Cunningham | 47.62       | Margaret       | Park          |
| Luis       | Rojas      | 46.62       | Steve          | Johnson       |
| Hugh       | O'Reilly   | 45.62       | Jane           | Peacock       |
| Ladislav   | Kovács     | 45.62       | Jane           | Peacock       |

Steve Johnson manages two of the top five spenders (Helena Holý and Luis Rojas), while Jane Peacock manages the other two of the remaining three — the top accounts are concentrated among a small group of reps rather than spread evenly.

## Q9. Which artists have the most tracks in our catalog, and how does that compare to their revenue?

Hint: JOIN Artist -> Album -> Track -> InvoiceLine, compare COUNT(tracks) vs SUM(revenue)

### SQL Query

```
SELECT
    Artist.Name AS artist,
    COUNT(DISTINCT Track.TrackId) AS total_tracks,
    COALESCE(SUM(InvoiceLine.UnitPrice * InvoiceLine.Quantity), 0) AS total_revenue,
    COALESCE(SUM(InvoiceLine.UnitPrice * InvoiceLine.Quantity), 0)
```

```

    / COUNT(DISTINCT Track.TrackId) AS revenue_per_track
FROM Artist
JOIN Album ON Artist.ArtistId = Album.ArtistId
JOIN Track ON Album.AlbumId = Track.AlbumId
LEFT JOIN InvoiceLine ON Track.TrackId = InvoiceLine.TrackId
GROUP BY Artist.ArtistId, Artist.Name
ORDER BY total_tracks DESC
LIMIT 10;

```

**Answer / Insight**

| Artist          | Tracks | Revenue  | Revenue/Track | Comment                                  |
|-----------------|--------|----------|---------------|------------------------------------------|
| Iron Maiden     | 213    | \$138.60 | \$0.65        | Largest catalog; high total revenue      |
| U2              | 135    | \$105.93 | \$0.78        | Large catalog; strong revenue            |
| Led Zeppelin    | 114    | \$86.13  | \$0.76        | Large catalog; strong revenue            |
| Metallica       | 112    | \$90.09  | \$0.80        | Strong revenue efficiency                |
| Deep Purple     | 92     | \$43.56  | \$0.47        | Large catalog; relatively low revenue    |
| Lost            | 92     | \$81.59  | \$0.89        | Strong revenue for catalog size          |
| Pearl Jam       | 67     | \$31.68  | \$0.47        | Moderate catalog; relatively low revenue |
| Lenny Kravitz   | 57     | \$25.74  | \$0.45        | Relatively weak revenue                  |
| Various Artists | 56     | \$28.71  | \$0.51        | Moderate catalog; low-moderate revenue   |
| The Office      | 53     | \$49.75  | \$0.94        | Small catalog; highest efficiency here   |

The two largest catalogs by track count (Iron Maiden, Deep Purple) don't necessarily have the best revenue efficiency — The Office has by far the smallest catalog here (53 tracks) but the highest revenue per track (\$0.94), while Deep Purple's 92 tracks earn only \$0.47 each. Catalog size alone doesn't predict sales performance.

## Tier 4 — Subqueries / CTEs (Management asks)

### Q10. Which customers spent above the average customer spend?

Hint: Subquery: WHERE total\_spend > (SELECT AVG(...))

**SQL Query**

```

SELECT
    Customer.CustomerId,
    Customer.FirstName,
    Customer.LastName,
    SUM(Invoice.Total) AS total_spend
FROM Customer
JOIN Invoice ON Customer.CustomerId = Invoice.CustomerId
GROUP BY Customer.CustomerId, Customer.FirstName, Customer.LastName
HAVING SUM(Invoice.Total) > (
    SELECT AVG(total_spend)
    FROM (
        SELECT SUM(Total) AS total_spend
        FROM Invoice
        GROUP BY CustomerId
    ) AS customer_spending
)
ORDER BY total_spend DESC;

```

**Answer / Insight**

| ID | Customer              | Total Spend |
|----|-----------------------|-------------|
| 6  | Helena Holý           | \$49.62     |
| 26 | Richard Cunningham    | \$47.62     |
| 57 | Luis Rojas            | \$46.62     |
| 45 | Ladislav Kovács       | \$45.62     |
| 46 | Hugh O'Reilly         | \$45.62     |
| 24 | Frank Ralston         | \$43.62     |
| 28 | Julia Barnett         | \$43.62     |
| 37 | Fynn Zimmermann       | \$43.62     |
| 7  | Astrid Gruber         | \$42.62     |
| 25 | Victor Stevens        | \$42.62     |
| 44 | Terhi Hämäläinen      | \$41.62     |
| 5  | František Wichterlová | \$40.62     |
| 43 | Isabelle Mercier      | \$40.62     |
| 48 | Johannes Van der Berg | \$40.62     |
| 1  | Luís Gonçalves        | \$39.62     |
| 3  | François Tremblay     | \$39.62     |
| 4  | Björn Hansen          | \$39.62     |
| 17 | Jack Smith            | \$39.62     |
| 20 | Dan Miller            | \$39.62     |
| 22 | Heather Leacock       | \$39.62     |
| 34 | João Fernandes        | \$39.62     |
| 42 | Wyatt Girard          | \$39.62     |

22 of the 59 customers (about 37%) spend above the customer average — the distribution is right-skewed, with a smaller group of above-average spenders and a longer tail of lower-spending customers.

**Q11. For each customer, what was their first purchase date and most recent purchase date?**

Hint: GROUP BY CustomerId, MIN(InvoiceDate), MAX(InvoiceDate)

**SQL Query**

```
SELECT
    CustomerId,
    MIN(InvoiceDate) AS first_purchase_date,
    MAX(InvoiceDate) AS most_recent_purchase_date
FROM Invoice
GROUP BY CustomerId
ORDER BY CustomerId;
```

**Answer / Insight**

| Customer ID | First Purchase | Most Recent Purchase |
|-------------|----------------|----------------------|
| 1           | 3/11/2022      | 8/7/2025             |
| 2           | 1/1/2021       | 7/13/2024            |

| Customer ID | First Purchase | Most Recent Purchase |
|-------------|----------------|----------------------|
| 3           | 3/11/2022      | 9/20/2025            |
| 4           | 1/2/2021       | 10/3/2025            |
| 5           | 12/8/2021      | 5/6/2025             |
| 6           | 7/11/2021      | 11/13/2025           |
| 7           | 12/8/2021      | 6/19/2025            |
| 8           | 1/3/2021       | 10/4/2025            |
| 9           | 9/6/2021       | 2/2/2025             |
| 10          | 4/9/2021       | 8/12/2025            |
| 11          | 9/6/2021       | 3/18/2025            |
| 12          | 5/23/2021      | 10/5/2025            |
| 13          | 6/5/2021       | 11/1/2024            |
| 14          | 1/6/2021       | 5/11/2025            |
| 15          | 6/5/2021       | 12/15/2024           |
| 16          | 2/19/2021      | 7/4/2025             |
| 17          | 3/4/2021       | 7/31/2024            |
| 18          | 5/12/2022      | 10/8/2025            |
| 19          | 3/4/2021       | 9/13/2024            |
| 20          | 5/12/2022      | 11/21/2025           |
| 21          | 3/5/2021       | 12/4/2025            |
| 22          | 2/8/2022       | 7/7/2025             |
| 23          | 1/11/2021      | 12/4/2025            |
| 24          | 2/8/2022       | 8/20/2025            |
| 25          | 3/6/2021       | 12/5/2025            |
| 26          | 11/7/2021      | 4/5/2025             |
| 27          | 6/10/2021      | 10/13/2025           |
| 28          | 11/7/2021      | 5/19/2025            |
| 29          | 7/24/2021      | 12/6/2025            |
| 30          | 8/6/2021       | 1/2/2025             |
| 31          | 3/9/2021       | 7/12/2025            |
| 32          | 8/6/2021       | 2/15/2025            |
| 33          | 4/22/2021      | 9/4/2025             |
| 34          | 5/5/2021       | 10/1/2024            |
| 35          | 7/13/2022      | 12/9/2025            |
| 36          | 5/5/2021       | 11/14/2024           |
| 37          | 1/19/2021      | 6/3/2025             |
| 38          | 2/1/2021       | 6/30/2024            |
| 39          | 4/11/2022      | 9/7/2025             |
| 40          | 2/1/2021       | 8/13/2024            |

| Customer ID | First Purchase | Most Recent Purchase |
|-------------|----------------|----------------------|
| 41          | 4/11/2022      | 10/21/2025           |
| 42          | 2/2/2021       | 11/3/2025            |
| 43          | 1/8/2022       | 6/6/2025             |
| 44          | 8/11/2021      | 12/14/2025           |
| 45          | 1/8/2022       | 7/20/2025            |
| 46          | 2/3/2021       | 11/4/2025            |
| 47          | 10/7/2021      | 3/5/2025             |
| 48          | 5/10/2021      | 9/12/2025            |
| 49          | 10/7/2021      | 4/18/2025            |
| 50          | 6/23/2021      | 11/5/2025            |
| 51          | 7/6/2021       | 12/2/2024            |
| 52          | 2/6/2021       | 6/11/2025            |
| 53          | 7/6/2021       | 1/15/2025            |
| 54          | 3/22/2021      | 8/4/2025             |
| 55          | 4/4/2021       | 8/31/2024            |
| 56          | 6/12/2022      | 11/8/2025            |
| 57          | 4/4/2021       | 10/14/2024           |
| 58          | 6/12/2022      | 12/22/2025           |
| 59          | 4/5/2021       | 5/30/2024            |

Nearly every customer has a first purchase dating back to 2021 or 2022 and a recent purchase within the last year or two, indicating strong long-term retention across the customer base rather than a lot of one-time buyers.

## Q12. Identify customers who bought in 2023 but not in 2024 (churn candidates).

Hint: LEFT JOIN 2024 purchases, WHERE 2024 side IS NULL

### SQL Query

```
SELECT DISTINCT
  c.CustomerId,
  c.FirstName,
  c.LastName
FROM Customer AS c
JOIN Invoice AS i2023
  ON c.CustomerId = i2023.CustomerId
  AND YEAR(i2023.InvoiceDate) = 2023
LEFT JOIN Invoice AS i2024
  ON c.CustomerId = i2024.CustomerId
  AND YEAR(i2024.InvoiceDate) = 2024
WHERE i2024.CustomerId IS NULL;
```

### Answer / Insight

| ID | First Name | Last Name |
|----|------------|-----------|
| 8  | Daan       | Peeters   |
| 12 | Roberto    | Almeida   |
| 14 | Mark       | Philips   |

| ID | First Name | Last Name |
|----|------------|-----------|
| 29 | Robert     | Brown     |
| 31 | Martha     | Silk      |
| 33 | Ellie      | Sullivan  |
| 35 | Madalena   | Sampaio   |
| 46 | Hugh       | O'Reilly  |
| 50 | Enrique    | Muñoz     |
| 52 | Emma       | Jones     |
| 54 | Steve      | Murray    |
| 56 | Diego      | Gutiérrez |

12 customers who purchased in 2023 did not return in 2024 — this is a small, specific list Marketing could target with a win-back campaign before they lapse further.

## Tier 5 — Window Functions (Senior analyst asks)

### Q13. Rank each employee by total sales revenue they're responsible for.

Hint: RANK() OVER (ORDER BY SUM(revenue) DESC)

#### SQL Query

```
SELECT
  e.EmployeeId,
  e.FirstName,
  e.LastName,
  SUM(i.Total) AS total_revenue,
  RANK() OVER (ORDER BY SUM(i.Total) DESC) AS sales_rank
FROM Employee AS e
JOIN Customer AS c ON e.EmployeeId = c.SupportRepId
JOIN Invoice AS i ON c.CustomerId = i.CustomerId
GROUP BY e.EmployeeId, e.FirstName, e.LastName
ORDER BY sales_rank;
```

#### Answer / Insight

| Emp. ID | First Name | Last Name | Total Revenue | Rank |
|---------|------------|-----------|---------------|------|
| 3       | Jane       | Peacock   | \$833.04      | 1    |
| 4       | Margaret   | Park      | \$775.40      | 2    |
| 5       | Steve      | Johnson   | \$720.16      | 3    |

Only three employees carry support-rep responsibility for customers. Jane Peacock leads with \$833.04 in attributed revenue, about 15% ahead of Margaret Park and 16% ahead of Steve Johnson — a fairly even three-way split at the top.

### Q14. For each customer, show a running total of their spend over time, ordered by invoice date.

Hint: SUM(Total) OVER (PARTITION BY CustomerId ORDER BY InvoiceDate)

#### SQL Query

```
SELECT
```

```

c.CustomerId,
CONCAT(c.FirstName, ' ', c.LastName) AS full_name,
i.InvoiceId,
i.InvoiceDate,
i.Total,
SUM(i.Total) OVER (
  PARTITION BY c.CustomerId
  ORDER BY i.InvoiceDate, i.InvoiceId
) AS running_spend
FROM Customer AS c
JOIN Invoice AS i ON c.CustomerId = i.CustomerId
ORDER BY c.CustomerId, i.InvoiceDate, i.InvoiceId;

```

### Answer / Insight

Sample output for Customer 1 (Luís Gonçalves) — the query returns one row per invoice for every customer; the first customer's rows are shown below as an example.

| Invoice ID | Invoice Date | Total | Running Spend |
|------------|--------------|-------|---------------|
| 98         | 3/11/2022    | 3.98  | 3.98          |
| 121        | 6/13/2022    | 3.96  | 7.94          |
| 143        | 9/15/2022    | 5.94  | 13.88         |
| 195        | 5/6/2023     | 0.99  | 14.87         |
| 316        | 10/27/2024   | 1.98  | 16.85         |
| 327        | 12/7/2024    | 13.86 | 30.71         |
| 382        | 8/7/2025     | 8.91  | 39.62         |

For this customer, spend accumulates steadily over four years, with a notable jump on 12/7/2024 (\$13.86 in one invoice) that nearly doubles their running total for the year.

## Q15. What is the month-over-month revenue growth rate?

Hint: LAG() OVER (ORDER BY month) to compare to prior month

### SQL Query

```

WITH monthly_revenue AS (
  SELECT
    DATE_FORMAT(InvoiceDate, '%Y-%m') AS month,
    SUM(Total) AS revenue
  FROM Invoice
  GROUP BY DATE_FORMAT(InvoiceDate, '%Y-%m')
)
SELECT
  month,
  revenue,
  LAG(revenue) OVER (ORDER BY month) AS previous_month_revenue,
  ROUND(
    (revenue - LAG(revenue) OVER (ORDER BY month))
    / LAG(revenue) OVER (ORDER BY month) * 100,
    2
  ) AS growth_rate
FROM monthly_revenue
ORDER BY month;

```

### Answer / Insight

| Month   | Revenue | Prev. Month Revenue | Growth Rate (%) |
|---------|---------|---------------------|-----------------|
| 2021-01 | 35.64   | —                   | —               |
| 2021-02 | 37.62   | 35.64               | 5.56            |
| 2021-03 | 37.62   | 37.62               | 0.00            |
| 2021-04 | 37.62   | 37.62               | 0.00            |
| 2021-05 | 37.62   | 37.62               | 0.00            |
| 2021-06 | 37.62   | 37.62               | 0.00            |
| 2021-07 | 37.62   | 37.62               | 0.00            |
| 2021-08 | 37.62   | 37.62               | 0.00            |
| 2021-09 | 37.62   | 37.62               | 0.00            |
| 2021-10 | 37.62   | 37.62               | 0.00            |
| 2021-11 | 37.62   | 37.62               | 0.00            |
| 2021-12 | 37.62   | 37.62               | 0.00            |
| 2022-01 | 52.62   | 37.62               | 39.87           |
| 2022-02 | 46.62   | 52.62               | -11.40          |
| 2022-03 | 44.62   | 46.62               | -4.29           |
| 2022-04 | 37.62   | 44.62               | -15.69          |
| 2022-05 | 37.62   | 37.62               | 0.00            |
| 2022-06 | 37.62   | 37.62               | 0.00            |
| 2022-07 | 37.62   | 37.62               | 0.00            |
| 2022-08 | 37.62   | 37.62               | 0.00            |
| 2022-09 | 36.63   | 37.62               | -2.63           |
| 2022-10 | 37.62   | 36.63               | 2.70            |
| 2022-11 | 37.62   | 37.62               | 0.00            |
| 2022-12 | 37.62   | 37.62               | 0.00            |
| 2023-01 | 37.62   | 37.62               | 0.00            |
| 2023-02 | 37.62   | 37.62               | 0.00            |
| 2023-03 | 37.62   | 37.62               | 0.00            |
| 2023-04 | 51.62   | 37.62               | 37.21           |
| 2023-05 | 42.62   | 51.62               | -17.44          |
| 2023-06 | 50.62   | 42.62               | 18.77           |
| 2023-07 | 37.62   | 50.62               | -25.68          |
| 2023-08 | 37.62   | 37.62               | 0.00            |
| 2023-09 | 37.62   | 37.62               | 0.00            |
| 2023-10 | 37.62   | 37.62               | 0.00            |
| 2023-11 | 23.76   | 37.62               | -36.84          |
| 2023-12 | 37.62   | 23.76               | 58.33           |
| 2024-01 | 37.62   | 37.62               | 0.00            |
| 2024-02 | 37.62   | 37.62               | 0.00            |

| Month   | Revenue | Prev. Month Revenue | Growth Rate (%) |
|---------|---------|---------------------|-----------------|
| 2024-03 | 37.62   | 37.62               | 0.00            |
| 2024-04 | 37.62   | 37.62               | 0.00            |
| 2024-05 | 37.62   | 37.62               | 0.00            |
| 2024-06 | 37.62   | 37.62               | 0.00            |
| 2024-07 | 39.62   | 37.62               | 5.32            |
| 2024-08 | 47.62   | 39.62               | 20.19           |
| 2024-09 | 46.71   | 47.62               | -1.91           |
| 2024-10 | 42.62   | 46.71               | -8.76           |
| 2024-11 | 37.62   | 42.62               | -11.73          |
| 2024-12 | 37.62   | 37.62               | 0.00            |
| 2025-01 | 37.62   | 37.62               | 0.00            |
| 2025-02 | 27.72   | 37.62               | -26.32          |
| 2025-03 | 37.62   | 27.72               | 35.71           |
| 2025-04 | 33.66   | 37.62               | -10.53          |
| 2025-05 | 37.62   | 33.66               | 11.76           |
| 2025-06 | 37.62   | 37.62               | 0.00            |
| 2025-07 | 37.62   | 37.62               | 0.00            |
| 2025-08 | 37.62   | 37.62               | 0.00            |
| 2025-09 | 37.62   | 37.62               | 0.00            |
| 2025-10 | 37.62   | 37.62               | 0.00            |
| 2025-11 | 49.62   | 37.62               | 31.90           |
| 2025-12 | 38.62   | 49.62               | -22.17          |

Most months show 0% growth (flat repeat revenue), punctuated by occasional swings of  $\pm 20\text{--}40\%$  — the sharpest drop is November 2023 (-36.84%) followed by the sharpest recovery the very next month (+58.33%), and the most recent data point (Dec 2025) shows a -22.17% dip worth monitoring.